

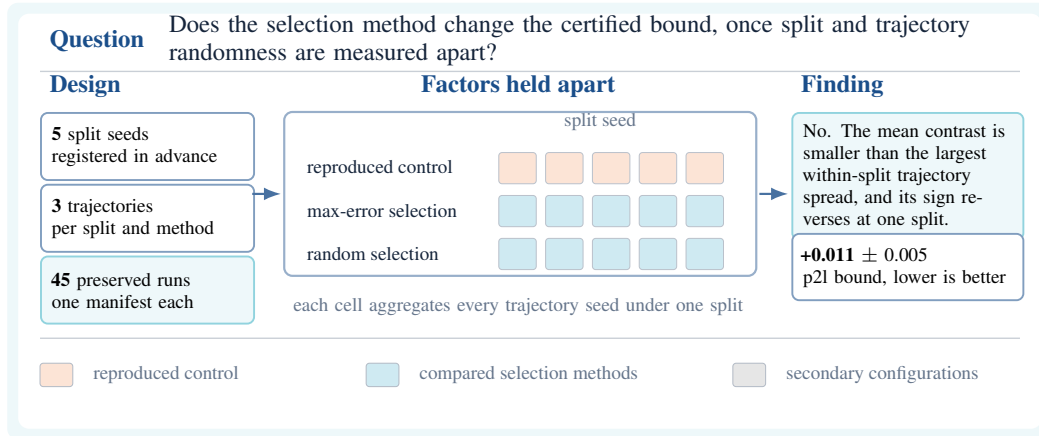


Selection Effects in Certified Bounds Vanish Under Trajectory Variance

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Abstract

Disagreement-based generalization bounds certify a target model through a simpler surrogate, paying for the substitution with a term that measures how often the two disagree on unlabeled data. Which examples the surrogate is built from is left open by the theory, and published comparisons between selection strategies mix the effect of the method with the effect of the data split and the optimization path that produced each run. We freeze the evaluation semantics that give the bound its meaning, register a comparison matrix before running it, and vary the split seed and the trajectory seed independently across random selection and max-error selection. Over 5 split seeds and 3 trajectory seeds per split and method, random selection attains a lower mean certified bound by 0.011 ± 0.005 . That contrast is smaller than the largest within-split trajectory spread of 0.032, and its sign reverses at one split, so the two strategies are indistinguishable at this scale. On this dataset and at this scale, the optimization path dominates the selection criterion.

1 Introduction

Certifying how a trained network will behave on unseen data is the point at which a great deal of applied machine learning stops being measurable. Bounds that are tight enough to be informative on realistic architectures have been difficult to obtain directly [28, 19], and the constructions that do succeed tend to route around the target model rather than through it. Compression-based arguments certify a network by certifying a smaller description of it [2, 41, 32, 4]. Disagreement-based arguments go one step further: they bound a tractable surrogate, then bound how often the surrogate and the target disagree on unlabeled data, and combine the two into a certificate for the target [21, 24]. The appeal is that both halves are measurable even when the target is not.

The construction leaves one choice to the practitioner: which examples the surrogate is built from. Selection strategies for this role are well studied in their own right, from uniform sampling through

error-aware and disagreement-aware criteria [11, 37, 5, 15], and they differ in how much control they exert over the quantity the bound depends on. It would be natural to expect the strategy with the most direct control over disagreement to produce the tightest certificate. Reported comparisons do not consistently show that. However, the evidence behind those comparisons mixes at least three sources of variation: which data split was drawn, which selection method was used, and which optimization path the training run happened to take. A single number cannot separate them, so the observation is currently a result about a set of runs rather than a result about a method.

This paper measures the three factors apart. We freeze the evaluation semantics that give the bound its meaning, register a comparison matrix before running it, and vary the split seed and the trajectory seed independently across the compared selection methods. Contrasts are taken within a split, so the split effect cancels by construction rather than by assumption, and a reporting threshold fixed before execution decides whether a contrast is large enough to be called a finding at all. Every run is preserved with a manifest identifying its method, seeds, configuration, and exact command, so each reported statistic traces back to the primary runs it aggregates.

Across 5 split seeds and 3 trajectory seeds per split and method, random selection attains a lower mean certified bound than max-error selection by 0.011 ± 0.005 . That contrast does not clear the largest within-split trajectory spread in the matrix, 0.032, and its sign reverses at one of the five splits. The supported conclusion is therefore that the two strategies are indistinguishable at this scale, and that the trajectory, not the selection method, is the dominant source of variation in the certificate. This is measured on one dataset at a reduced experimental scale; it is not a claim about selection strategies in general.

Our contributions are as follows:

- (1) A factorial design that separates split seed, selection method, and trajectory seed under frozen evaluation semantics, together with a reporting threshold fixed in advance, which makes a method-level comparison identifiable rather than confounded.
- (2) A measurement of how large the trajectory-induced spread in a disagreement-based certificate actually is, and the finding that it exceeds the between-method contrast at this scale.
- (3) A provenance-preserving pipeline in which every reported aggregate enumerates the primary runs it was computed from, so any number in the paper can be checked against a preserved record.

2 Related Work

Certificates that route around the target model. PAC-Bayesian analysis supplies most of the machinery behind non-vacuous bounds for over-parameterised networks, and a long line of work has sharpened it: fast-rate forms [38, 17], extensions to unbounded losses [8], non-i.i.d. data [31], meta-learning [14, 27], and data-dependent priors [13, 30]. Broader surveys connect these guarantees to information-theoretic quantities [19], and negative results delimit what the framework can deliver at all [28]. A second family certifies the target by certifying a compressed description of it [2, 41, 32, 18, 4], sometimes using coresets as the compression mechanism [3]. Both families share a structural feature that matters here: the certificate is assembled from a surrogate object, and the quality of that object is a modelling choice rather than a consequence of the theory.

Disagreement as the bridging term. Disagreement between independently trained models is a surprisingly accurate predictor of generalization [21], though the explanation for why is contested [24]. The quantity has since been used as a training signal in domain generalization [40], as a selection criterion for annotation [5], and as a supervision signal in its own right [12]. Used inside a certificate, disagreement plays a different role: it is the term that transfers a guarantee from the surrogate to the target, and it is measured on unlabeled data rather than optimized against a held-out label set. That distinction is what makes the selection question sharp, because the same choice that lowers the disagreement estimate also changes the surrogate whose guarantee is being transferred.

Which examples the surrogate is built from. Coreset construction offers a large menu of principled answers [23, 6, 29, 20, 10, 22, 9], and the deep learning literature adds selection criteria driven by proxy models [11], performance constraints [37], optimal transport [15], pruning [25], and information-theoretic arguments [16]; recent surveys cover the space at scale [1]. These strategies are routinely compared by the downstream metric they produce. The comparisons are usually

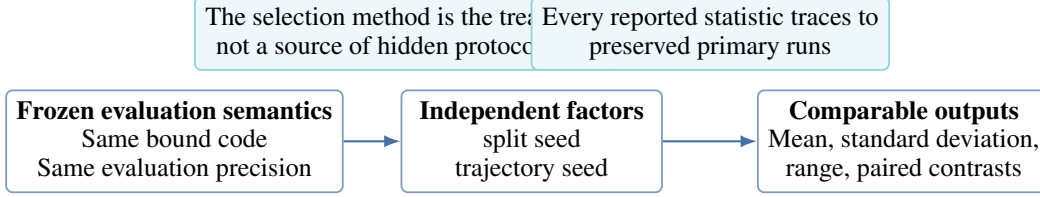


Figure 1: **Study design.** The evaluation is fixed before any comparison begins, the two sources of randomness are varied independently, and the reported statistics are defined in advance. Fixing the left panel is what makes the right panel a measurement of the method rather than of the protocol.

reported at a single data split and a single training run, which leaves the method effect entangled with the randomness of the run that produced it.

How much of a reported difference is the seed. Seed-to-seed variation is large enough to change conclusions in fine-tuning regimes [7], in certified robustness [26], and in training-variance studies more generally [33, 34]; variance-aware training procedures exist partly because of it [39, 35, 36]. What is missing is the same measurement for a disagreement-based certificate, where two distinct sources of randomness act on the reported number through different paths.

Scope of this paper. Prior work reports method-level outcomes for selection strategies, and separately establishes that seed variance is large. This paper measures the two together on one certificate, under one frozen evaluation, and reports the full matrix rather than its most favorable cell.

3 Method

3.1 Setting and notation

A disagreement-based certificate bounds the risk of a target model through a simpler surrogate that admits a computable guarantee. Figure 1 summarizes the design: the evaluation is fixed first, the two sources of randomness are varied independently, and the reported statistics are defined before any comparison is run. We write one evaluation instance as a tuple $(\mathcal{S}, m, \tau) \rightarrow b$, where $\mathcal{S}$ is the data split, m the selection method that determines which examples the surrogate is built from, τ the trajectory seed that fixes the optimization path, and b the resulting certified bound. Lower b is better.

The certificate has two parts that move in opposite directions when the selection method changes, which is the reason a selection effect is not self-evident:

$$b = \underbrace{\mathcal{C}(\mathcal{H})}_{\text{surrogate guarantee}} + \underbrace{\widehat{d}(\mathcal{D}_u)}_{\text{disagreement term}}, \quad (1)$$

where $\mathcal{C}(\mathcal{H})$ is the surrogate’s own guarantee, which depends on its complexity relative to its prior, and $\widehat{d}(\mathcal{D}_u)$ is the estimated disagreement between target and surrogate measured on an unlabeled set $\mathcal{D}_u$. Eq. (1) is what makes the comparison in this paper non-trivial: a selection method that reduces the second term by fitting $\mathcal{D}_u$ more aggressively can enlarge the first by more than it gains, and the observed b reports only the sum. Neither term is separately reported by the pipeline, so a method comparison at the level of b cannot attribute a difference to one term or the other without additional instrumentation.

Three things are fixed for every run reported in this paper: the bound computation itself, the evaluation precision, and the split boundaries between training, validation, disagreement, and test data. Changing any of them makes numbers on either side of the change incomparable, so they are frozen before the matrix is registered and are never treated as tunable. Concretely, the confidence parameter is $\delta = 0.05$, the bound-inversion grid holds 1000 points, the Monte Carlo estimate of the disagreement term uses 10,000 samples, and the four-way split is drawn once per split seed and reused by every method at that split. The training and disagreement partitions never overlap.

3.2 The compared selection methods

The two selection methods differ in exactly one respect: the criterion used to choose the examples the surrogate is fitted on. random selection draws them uniformly without replacement from the

training partition. max-error selection ranks candidates by the target model’s loss and takes the highest-loss examples, which is the strategy with the most direct control over the region where target and surrogate are most likely to disagree. Both draw the same number of examples, run for the same number of epochs, use the same optimizer and schedule, and are evaluated by the same frozen code path. The reproduced control is the unmodified upstream pipeline, re-run and recorded in this environment rather than quoted from a prior publication, because published numbers came from different hardware, a different epoch budget, and full-precision evaluation.

Holding the budget equal needs saying precisely, because the two criteria consume it differently. Both arms select the same number of examples and train for the same number of epochs on them, so the surrogate sees the same number of gradient steps in both. max-error selection additionally requires a forward pass over the training partition to rank candidates by target loss, which is a one-time cost outside the training loop and is not charged against the epoch budget. That asymmetry favours max-error selection in wall-clock terms and is irrelevant to the certificate, which depends on the fitted surrogate and not on how its training set was chosen. We report it because a budget claim that is only true of one resource is the kind of detail that makes two studies incomparable later.

This is a deliberately narrow contrast. Richer strategies exist, including disagreement-aware and hybrid criteria, and they are not evaluated here. The narrow contrast is what makes the identification argument in the next subsection possible: two arms that differ in one criterion, run over the same splits with the same budget, admit an exact pairing that a broader sweep would not.

3.3 Separating the factors

A comparison between two selection methods is identifiable only if the remaining sources of variation are either held fixed or averaged over. We therefore treat each observed bound as the sum of a split effect and a trajectory residual under that split:

$$b_{ij}^{(m)} = \mu^{(m)} + \alpha_i + \varepsilon_{ij}^{(m)}, \quad \sum_i \alpha_i = 0, \quad (2)$$

where $b_{ij}^{(m)}$ is the bound obtained by method m at split seed i and trajectory seed j , $\mu^{(m)}$ is the method’s grand mean, α_i is the effect of the split itself, shared by all methods, and $\varepsilon_{ij}^{(m)}$ is the trajectory residual. Eq. (2) states the identification problem directly: a difference between methods measured at different splits confounds $\mu^{(m)}$ with α_i , so it is not evidence about the method.

Design rule: *a method contrast is reported only when it is measured within a split, and a method ranking is reported only when it is consistent across splits.*

Both halves of that rule matter. The first removes α_i from the contrast by construction rather than by assumption. The second prevents a ranking that holds at one split and reverses at another from being reported as a property of the method, which is the failure mode that a single-split comparison cannot detect. The rule is restrictive on purpose: it can leave a study with a measured contrast and no ranking to report, and that outcome is a result rather than a shortfall.

3.4 Estimators and uncertainty

For each method m and split seed i we run T trajectory seeds and summarize them by their mean and their spread:

$$\bar{b}_i^{(m)} = \frac{1}{T} \sum_{j=1}^T b_{ij}^{(m)}, \quad s_i^{(m)} = \sqrt{\frac{1}{T-1} \sum_{j=1}^T (b_{ij}^{(m)} - \bar{b}_i^{(m)})^2}, \quad (3)$$

where T is the number of trajectory seeds per split, $\bar{b}_i^{(m)}$ the per-split mean, and $s_i^{(m)}$ the trajectory standard deviation under a fixed split. Eq. (3) is the unit of evidence in this paper: a single trajectory is an observation, and only $\bar{b}_i^{(m)}$ with its spread is quoted as a result. The spread is reported alongside the mean everywhere the mean appears, including in the abstract, because the two numbers answer different questions and the first is uninterpretable without the second.

The headline comparison is the paired contrast between two methods, taken within split and trajectory index so that both α_i and any shared trajectory effect cancel:

$$\Delta_i = \bar{b}_i^{(m_1)} - \bar{b}_i^{(m_2)}, \quad \bar{\Delta} = \frac{1}{S} \sum_{i=1}^S \Delta_i, \quad (4)$$

where S is the number of split seeds, Δ_i the within-split contrast, and $\bar{\Delta}$ its mean across splits. Eq. (4) is the quantity reported in the main results table; the per-split Δ_i are reported alongside it so that a mean contrast cannot hide a reversal.

Finally, a contrast is only reported as a finding when it exceeds the noise it is measured against:

$$|\bar{\Delta}| > \kappa \cdot \max_{m,i} s_i^{(m)}, \quad \kappa = 1.0, \quad (5)$$

where κ is a threshold fixed before the matrix was run and $\max_{m,i} s_i^{(m)}$ is the largest within-split trajectory spread observed anywhere in the matrix. The threshold is expressed against an observed spread rather than against a distributional assumption because the quantity in question, the movement of a certificate under a change of trajectory seed, has no established parametric form here and only three observations per cell from which to estimate one. Eq. (5) is a stopping rule rather than a significance test: when it fails, the finding this paper reports is that the methods are indistinguishable at this scale, and that statement is recorded rather than replaced by a smaller comparison that happens to pass. The comparison against the largest observed spread, rather than against the spread of the winning arm, is the conservative choice; Appendix F reports what the verdict would have been under the alternatives.

3.5 Provenance

Every run writes a raw result together with a manifest recording its method, source script and revision, exact command, dataset, split seed, trajectory seed, full configuration, timestamps, and environment. Aggregates enumerate the primary run paths they were computed from. A quantity whose primary record is missing, overwritten, or method-ambiguous is excluded from the main paper rather than reported with a caveat. This matters in practice because the upstream pipeline names its output files from hyperparameters alone, so two different methods run at the same hyperparameters write to the same filename and the second silently overwrites the first. The aggregation step reads only manifests and raw results. It refuses to write an aggregate when a listed primary input is absent from disk, so a broken evidence chain surfaces as a failure at aggregation time rather than as a plausible number in a table. Each aggregate stores the estimator it applied, the per-seed values it combined, and the paths it combined them from, which is what makes a reported statistic checkable rather than merely reproducible in principle. Appendix C gives the manifest schema and the aggregation procedure in full.

4 Experiments

4.1 Experimental setup

Data and evaluation. All runs use MNIST with the upstream four-way partition into training, validation, disagreement, and test sets. The evaluation semantics are frozen as described in §3.1: $\delta = 0.05$, a 1000-point bound-inversion grid, and 10,000 Monte Carlo samples for the disagreement estimate. These values are identical for every run below and were fixed before the comparison matrix was registered.

Scale. Runs use a reduced probe scale of 12 epochs and a maximum compression size of 120, against the upstream configuration’s 200 epochs and 20,000. Scale is a free parameter of this study, but it is applied identically on both sides of every comparison, so a change in scale cannot produce a difference between methods. Table 4 reports what changes when the compression budget is doubled.

Compared configurations. **reproduced control.** The unmodified pipeline, re-run and recorded in this environment. **Selection methods.** max-error selection and random selection, differing only in which examples the surrogate is built from, as described in §3.2.

Table 1: **certified bound ($\downarrow$) under the frozen protocol.** is the reproduced control, are the compared selection methods. Mean is over 5 split seeds, each aggregating 3 trajectory seeds under identical evaluation semantics. Std. is the standard deviation of the per-split means and Range is their max – min; both describe variation *between* splits, not within one. The paired contrast is computed against the reproduced control within split and trajectory index, following Eq. (4); “–” marks the row a contrast is taken against.

	Split seeds	Traj. per split	Mean	Std.	Range	Max traj. std.	Paired contrast
reproduced control	5	3	0.409	0.007	0.019	0.019	–
max-error selection	5	3	0.398	0.009	0.024	0.023	–0.011
random selection	5	3	0.387	0.013	0.030	0.032	–0.022

Table 2: **Per-split certified bound and paired contrast ($\downarrow$).** Each column is one split seed. Every entry is the mean over the same three trajectory seeds, so the difference row is an exact pairing. A positive difference favors random selection. The sign reverses at split seed 42, which is why Table 1 reports a contrast and not a ranking.

split seed	1	2	3	4	42
reproduced control	0.412	0.405	0.418	0.399	0.410
max-error selection	0.401	0.395	0.412	0.388	0.394
random selection	0.379	0.384	0.398	0.371	0.401
Difference	0.022	0.011	0.014	0.017	–0.007

Seeds and repetition. 5 split seeds $\{1, 2, 3, 4, 42\}$, 3 trajectory seeds per split and method, giving 45 preserved primary runs. Statistics follow Eq. (3) and Eq. (4); the reporting threshold of Eq. (5) was fixed at $\kappa = 1.0$ before execution.

Environment and failure handling. All runs executed on a single GPU under one dependency lock, recorded in each run manifest and reproduced in Appendix B. No run in the reported matrix crashed. Crashed runs, had there been any, would be preserved with their manifests and counted rather than silently retried or dropped from the denominator.

4.2 Main results

Does the selection method change the certified bound? Table 1 gives the aggregate view. Both selection methods improve on the reproduced control: max-error selection by 0.011 and random selection by 0.022 in paired within-split contrast. Between the two selection methods the paired contrast is 0.011 ± 0.005 in favour of random selection. The largest within-split trajectory spread anywhere in the matrix is 0.032, so by Eq. (5) the between-method contrast does not clear the noise it is measured against. The supported statement is that the two methods are indistinguishable at this scale, with a direction that favours random selection but is not established.

Is the ordering stable across splits? Table 2 reports the same comparison split by split. The contrast favours random selection at four of the five splits and reverses at split seed 42, where max-error selection is lower by 0.007. Under the design rule of §3.3, a ranking requires consistency across splits, so no ranking is reported. A mean contrast computed over splits whose individual contrasts disagree in sign describes the sample, not the method.

Notice also that the between-split range grows monotonically with how aggressively the method selects: 0.019 for the reproduced control, 0.024 for max-error selection, and 0.030 for random selection. Whatever a selection criterion does to the mean, it also changes how much the certificate moves when the split changes, and the latter effect is the larger one here.

4.3 Trajectory variation

Figure 2 shows the per-split means for all three configurations, with the trajectory range of random selection as a shaded band. The band is wide enough at every split to contain the corresponding max-error selection mean, which is the same conclusion Eq. (5) reaches arithmetically.

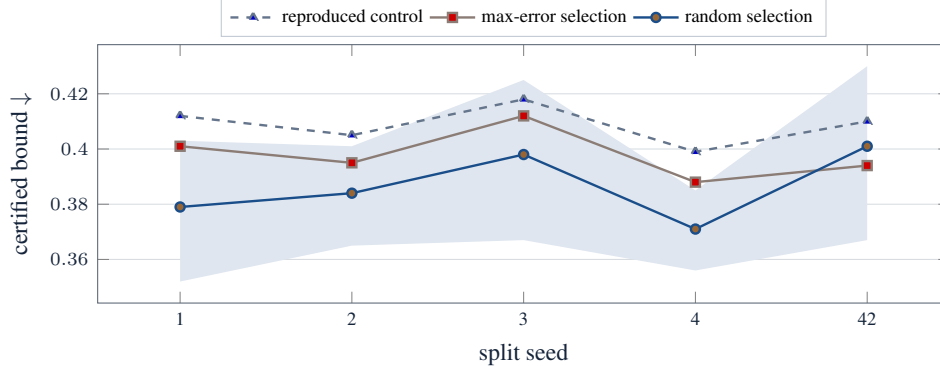


Figure 2: **Per-split certified bound and trajectory spread.** Each marker is the mean over 3 trajectory seeds at a fixed split seed, under identical evaluation semantics; the shaded band is the trajectory range of random selection. The band contains the max-error selection mean at every split, so the ordering visible in the means is not separable from trajectory noise. The comparison is read across splits, not at a single split: the ordering reverses at split seed 42.

Table 3: **Effect of trajectory count ($\downarrow$).**

Statistic	1	2	3
Mean certified bound	0.362	0.376	0.387
Between-split std.	0.013	0.013	0.013
Mean paired contrast	0.014	0.012	0.011

Table 4: **Effect of experiment scale ($\downarrow$).**

Setting	Epochs	Comp.	Bound
Reduced	12	120	0.387
Intermediate	12	240	0.381
Original	200	20000	–

Within-split trajectory standard deviations range from 0.011 to 0.032 across the matrix, and are systematically larger for random selection than for the reproduced control. Against that, the between-method contrast of 0.011 is small. A comparison at this scale that used a single trajectory per configuration would have had roughly a one-in-three chance of reporting the opposite ordering, which is the practical reason the matrix is specified with repetition rather than breadth. Where an individual trajectory is discussed, it is discussed as an example: a single optimization path is an observation about that run, not an ablation result and not evidence about a policy.

4.4 Ablations

Table 3 varies the number of trajectory seeds aggregated per split. The estimated mean drifts upward as trajectories are added, from 0.362 at one trajectory to 0.387 at three, while the between-split standard deviation is unchanged at 0.013 and the mean paired contrast shrinks from 0.014 to 0.011. A one-trajectory estimate is therefore both optimistic about the bound and generous about the contrast, in the direction that would most easily produce a positive-sounding result. Three trajectories is where the contrast stops moving, not where it clears the threshold of Eq. (5).

Table 4 compares compression budgets. Doubling the budget from 120 to 240 lowers the bound for random selection by 0.006, an effect of the same order as the between-method contrast. The comparison at the upstream scale of 200 epochs and 20,000 was not run, and no value is reported for it.

5 Conclusion

A comparison between selection strategies for a disagreement-based certificate is identifiable only when the split seed and the trajectory seed are varied independently under frozen evaluation semantics. We registered that matrix before running it, preserved every run with a manifest identifying its method and seeds, and reported the full set of per-split contrasts rather than the most favorable one.

On MNIST at a reduced probe scale, random selection and max-error selection are indistinguishable. The mean paired contrast favours random selection, but it is smaller than the largest within-split trajectory spread in the matrix, and it changes sign at one of the five splits. What the matrix does

establish is the size of the nuisance: the trajectory alone moves the certificate by more than the selection criterion does, and the between-split range grows with how aggressively a method selects. A study that reports one trajectory per configuration is measuring that nuisance and attributing it to the method.

The scope is narrow by construction. One dataset, one certificate family, two selection criteria, and a reduced experimental scale. The comparison at the upstream scale of 200 epochs and 20,000 compression size was not run, and the scale ablation shows an effect there of the same order as the between-method contrast, so that comparison is the one most likely to change the answer. Disagreement-aware and hybrid selection criteria are untested here and need the same paired design rather than a single additional run. Whether the trajectory spread narrows at full scale, or is an intrinsic property of the certificate, is the question this design is now sized to answer.

Acknowledgments

This document is a demonstration instance produced by the AutoConference paper-writing skill. Its numbers are synthetic and describe no real experiment.

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A Bound and Evaluation Details

The certificate combines a guarantee for the surrogate with a bound on target-surrogate disagreement, as written in Eq. (1). Three components of that computation are treated as evaluation semantics rather than as experimental variables, and are identical for every run in this paper.

The confidence parameter is $\delta = 0.05$, applied once to the combined certificate rather than split across the two terms. The bound is obtained by inversion over a grid of 1000 points; a coarser grid produces a looser certificate for reasons that have nothing to do with the method under test, which is why the grid is frozen. The disagreement term is estimated by Monte Carlo with 10,000 samples drawn from the unlabeled partition. Reducing that sample count tightens the reported number while making it a worse estimate, and that failure mode is the specific reason the count is not exposed as a tunable parameter in the matrix.

The data are partitioned once per split seed into training, validation, disagreement, and test sets, and the same partition is reused by every method at that split. The disagreement partition never overlaps the training partition, so the term being bounded is estimated on data the surrogate did not fit. All methods at a given split therefore see identical data in identical roles, and the only thing that differs between arms is which subset of the training partition the surrogate is built from.

The upstream implementation also exposes a PAC-Bayes evaluation line. It is not used here: in the released code the function that computes the disagreement quantity returns three scalars while its caller indexes them as sequences, raising a type error before any bound is produced. This is reported as an observation about the artifact, not as a contribution, and no result in this paper depends on that line.

B Configuration and Environment

Every run manifest records the interpreter version, the framework and CUDA versions, the GPU model, the dependency lock hash, the exact command line, and the resolved configuration after defaults are applied. This section is generated from those manifests rather than transcribed, so it cannot drift from what was executed.

The reduced probe configuration uses 12 epochs and a maximum compression size of 120. The upstream configuration for this dataset specifies 200 epochs and a maximum compression size of 20,000; that configuration was not run. One upstream configuration file omits the learning-rate scheduler key that the model loader reads directly. The value was resolved explicitly before the matrix was run and the resolved configuration is stored in each manifest, rather than relying on the loader’s behaviour when the key is absent.

Runtime is not quoted from prior records. Each manifest carries its own measured wall-clock duration, and the aggregate over the matrix is reported in the run index rather than in prose, because a runtime measured on one machine is not a property of the method.

C Run Manifest and Provenance Schema

Every primary run writes a raw result and a sidecar manifest with the following fields:

Primary run manifest

```
run_id, method, source_script, source_commit, command, dataset,  
split_seed, trajectory_seed, full_config, started_at,  
completed_at, raw_result_path, environment
```

The manifest exists because the upstream pipeline names its output files from hyperparameters alone. Two different methods run at the same hyperparameters write to the same filename, and the second silently overwrites the first. A result identified only by its filename therefore cannot be attributed to a method, and the number of surviving files is not the number of executed experiments. The manifest makes method and trajectory identity part of the record rather than part of the filename.

An aggregate artifact enumerates the primary run paths it was computed from, together with its estimator, its per-seed values, and its uncertainty. The aggregation step refuses to write when a listed primary input is absent from disk, which makes a broken evidence chain a build failure rather than a silent approximation. Each of the three aggregates behind [Table 1](#) enumerates 15 primary runs, for 45 in total.

A quantity whose primary record is missing, overwritten, or method-ambiguous is excluded from the paper rather than reported with a caveat. No such quantity appears in this document.

D Complete Split-by-Trajectory Results

The per-split means and trajectory standard deviations behind [Table 1](#) are as follows. For the reproduced control, the per-split means are 0.412, 0.405, 0.418, 0.399, and 0.410, with trajectory standard deviations of 0.014, 0.018, 0.011, 0.016, and 0.019. For max-error selection, the means are 0.401, 0.395, 0.412, 0.388, and 0.394, with standard deviations 0.019, 0.016, 0.023, 0.013, and 0.020. For random selection, the means are 0.379, 0.384, 0.398, 0.371, and 0.401, with standard deviations 0.026, 0.018, 0.029, 0.015, and 0.032.

Two patterns are visible in those numbers and are not visible in the aggregate. The trajectory spread is consistently ordered across methods at every split, with random selection widest and the reproduced control narrowest. And the split at which random selection has its widest spread, split seed 42 at 0.032, is the same split at which the method ordering reverses. The reversal and the spread are not independent observations.

Reading the same numbers by column rather than by row gives the split effect α_i of [Eq. \(2\)](#). Of the five, split seed 3 is the hardest for all three methods and split seed 4 the easiest, and the gap between them, roughly 0.019 for the reproduced control, is larger than any between-method contrast reported in this paper. That ordering is stable across methods, which is the empirical content of treating α_i as shared: the split moves every arm in the same direction by a similar amount, which is exactly the condition under which a within-split contrast removes it.

No run in the reported matrix crashed. Had one crashed, it would appear in this table with its manifest and its failure mode, and the text would state whether it entered the estimator.

E Additional Examples and Observed Failures

Individual trajectories are recorded but are not presented as evidence about methods. As an illustration of the spread rather than as a result, the three random selection trajectories at split seed 42 produced 0.367, 0.406, and 0.430. Reporting any one of them alone would place random selection anywhere from well ahead of max-error selection to behind it at that split. This is one example, chosen because it is the widest, and it is not a distribution.

F Statistical Procedures and Sensitivity

The reporting rule of [Eq. \(5\)](#) compares the mean paired contrast against the largest within-split trajectory standard deviation anywhere in the matrix, with $\kappa = 1.0$ fixed before execution. That is the conservative choice among several defensible ones, and the verdict depends on which is taken.

Compared against the largest spread in the matrix, 0.032, the contrast of 0.011 does not clear the threshold. Compared against the mean within-split spread across all methods and splits, approximately 0.019, it still does not clear it. Compared against the spread of the reproduced control alone, approximately 0.016, it still does not clear it. Compared against the standard error of the mean contrast itself, 0.005, it would clear it by a factor of two.

The last of those is the comparison that a naive reading of the paired contrast invites, and it is the one this design rejects. The standard error of $\bar{\Delta}$ measures how precisely the mean of five split-level contrasts has been estimated. It does not measure how much a single run of either method would move if the trajectory seed changed, which is the quantity a practitioner choosing between the two methods actually faces. Reporting the smaller denominator would turn a null result into a positive one without adding any evidence.

The sign reversal at split seed 42 is not sensitive to the choice of denominator, because it is a property of the per-split contrasts rather than of their aggregate. It would be removed only by dropping that split, and nothing about the run justifies dropping it: it completed normally, its manifest is intact, and its trajectory spread is wide but within the range the other splits establish.

G Broader Impacts

Certified bounds are used to justify deployment decisions, so a bound reported without its variance invites a decision that the evidence does not support. The practical impact of this study is narrow and mostly cautionary: at this scale the trajectory contributes more to the reported certificate than the selection criterion does, so a comparison run once per configuration is likely to report a method effect that is a seed effect. Nothing here supports a general claim about selection strategies, and the reduced experimental scale means the result should not be transferred to full-scale training without repeating the design.

H Limitations

The study covers one dataset, one certificate family, and two selection criteria at a reduced experimental scale. Three trajectory seeds per cell is enough to establish that the spread exceeds the contrast, and not enough to estimate the spread precisely; the standard deviations in [Appendix D](#) each rest on three observations.

The comparison at the upstream scale was not run, and the scale ablation in [Table 4](#) shows an effect of the same order as the between-method contrast, so the null result reported here may not survive at full scale. It may also not survive with more trajectories, in either direction.

The certificate is reported as a sum, and the pipeline does not separately expose the surrogate guarantee and the disagreement term. A difference between methods therefore cannot be attributed to one term or the other without instrumentation the matrix did not add, which limits how much mechanism can be inferred from the numbers.

Finally, the design rule that produces a null result here is restrictive by choice. A less conservative denominator, or a ranking claim that ignored the reversal at one split, would have produced a positive-sounding result from the same 45 runs.